

Doron Haviv

Research Scientist at Genentech

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EMPLOYMENT

Genentech BRAID - Biology Research — AI Development

May 2025 – Present

Research Scientist - Spatial Genomics

EDUCATION

Cornell University Joint Memorial Sloan Kettering Cancer Center

July 2019 – April 2025

Ph.D. in Computational Biology and Medicine

Research Advisor: Dana Pe'er

Thesis: From Cells to Niches - The Geometry of Spatial Transcriptomics

Technion - Israel Institute of Technology

October 2014 – October 2018

B.Sc. Electrical Engineering (*cum laude*)

B.Sc. Physics (*cum laude*)

Research Advisor: Omri Barak

Thesis: Understanding and Controlling Memory in Recurrent Neural Networks.

RESEARCH INTERESTS

generative models, single-cell genomics, optimal transport, spatial transcriptomics

PUBLICATIONS AND PRE-PRINTS

Haviv, D.*, De Brouwer, E.*, Rishabh, A., Ying, R., Scalia, G., Corrada-Bravo, H. 2026. When Riemann flows with Wasserstein: Generative Modeling of Probability Distributions on Manifolds. *In Review*

Andersson, A.*, Ismail, A.A.*, De Brouwer, E.*, Haviv, D.*, Biancalani, T., Cho, K., Scalia, G., BenTaieb, A., Corrada-Bravo, H. 2026. scCBGM: Single-Cell Editing via Concept Bottlenecks. Proceedings of the 43rd International Conference on Machine Learning, PMLR.

Haviv, D.*, Pooladian, A.A.*, Pe'er, D., Amos, B. 2025. Wasserstein Flow Matching: Generative modeling over families of distributions. Proceedings of the 42st International Conference on Machine Learning, PMLR.

Haviv, D., Kunes, R.Z., Dougherty, T. Burdziak, C., Nawy T., Gilbert A., Pe'er, D. 2024. Wasserstein Wormhole: Scalable Optimal Transport Distance with Transformers. Proceedings of the 41st International Conference on Machine Learning, PMLR.

Haviv, D., Remšík, J., Gatie, M., Snopkowski, C., Takizawa, M., Pereira, N., ..., Pe'er, D. 2024. The covariance environment defines cellular niches for spatial inference. Nature Biotechnology.

Highlighted in *Nature Biotechnology Research Briefing

Mani, S.*, Haviv, D.*, Kunes, R., Pe'er, D. 2022. SPOT: Spatial Optimal Transport for Analyzing Cellular Microenvironments. In NeurIPS 2022 Workshop on Learning Meaningful Representations of Life.

***Spotlight Presentation**

Elad, A.*, Haviv, D.*, Blau, Y. and Michaeli, T., 2019. Direct validation of the information bottleneck principle for deep nets. In Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops.

***Best poster award Statistical Deep Learning in Computer Vision Workshop**

Haviv, D., Rivkind, A. and Barak, O., 2019. Understanding and controlling memory in recurrent neural networks. Proceedings of the 36th International Conference on Machine Learning, PMLR.

Kunes, R.Z., Yin, M., Land, M., Haviv, D., Pe'er, D. and Tavaré, S., 2023, June. Gradient estimation for binary latent variables via gradient variance clipping. In Proceedings of the AAAI Conference on Artificial Intelligence.

Burdziak, C.*, Zhao, C. J.*, Haviv, D., Alonso-Curbelo, D., Lowe, S. W., Pe'er, D. 2023. scKINETICS: inference of regulatory velocity with single-cell transcriptomics data. Bioinformatics.

***Best paper award Intelligent Systems for Molecular Biology (ISMB) 2023.**

Burdziak, C.*, Alonso-Curbelo, D.*, Walle, T., Reyes, J., Barriga, F. M., Haviv, D., Xie, Y., Zhao, Z., Zhao, C. J., Chen, H.-A., Chaudhary, O., Masilionis, I., Choo, Z.-N., Gao, V., Luan, W., Wuest, A., Ho, Y.-J., Wei, Y., Quail, D. F., ... Pe'er, D. 2023. Epigenetic plasticity cooperates with cell-cell interactions to direct pancreatic tumorigenesis. Science.

***Highlighted in *Cancer Discovery*, *Nature Reviews Gastroenterology and Hepatology*, *Cell Trends in Cancer*, and EACR Highlights in Cancer Research.**

Raayoni, G., Gottlieb, S., Manor, Y., Pisha, G., Harris, Y., Mendlovic, U., Haviv, D., Hadad, Y. and Kaminer, I., 2021. Generating conjectures on fundamental constants with the Ramanujan Machine. Nature.

Highlighted in *New Scientist**AWARDS AND HONORS**

Best Poster Award , SDL-VC workshop, International Conference on Computer Vision	2019
Yehoraz Kasher Prize Best Student Project in Electrical Engineering, 3rd Place	2018
Technion - Israel Institute Of Technology President's List	2018
Technion - Israel Institute Of Technology Dean's List	2017
Technion - Israel Institute Of Technology Dean's List	2016

INVITED AND CONTRIBUTED TALKS

<i>Computational Genomics</i> class, Columbia University, New York, New York, USA	2026
Guest Lecture: <i>Spatial Transcriptomics - Technologies and Algorithms</i>	
Apple Machine Learning Research Group, Virtual	2024
Invited Talk: <i>Wasserstein Wormhole: Scalable Optimal Transport Distance with Transformers.</i>	
scverse Community Meeting, Virtual	2024
Invited Talk: <i>Reconstructing spatial context for single cell transcriptomics with ENVI</i>	
Department of Computer Science Colloquium, Columbia University, New York, New York, USA	2024
Invited Talk: <i>Reconstructing spatial context for single cell transcriptomics with ENVI</i>	
Ido Amit Lab, Weizmann Institute of Science, Rehovot, Israel	2024
Invited Talk: <i>Reconstructing spatial context for single cell transcriptomics with ENVI</i>	

10x Spatial World Tour, New York Genome Center, New York, New York, USA Invited Talk: <i>Reconstructing spatial context for single cell transcriptomics with ENVI</i>	2023
The Jackson Laboratory for Genomic Medicine, Farmington, Connecticut, USA Invited Talk: <i>Reconstructing spatial context for single cell transcriptomics with ENVI</i>	2022
Fusion Conference on Probing Human Disease using Single-Cell Technologies, Cancun, MX Contributed Talk: <i>Spatial Context of Heterogenous T Cell Response to Fungal Insult.</i>	2022
International conference on machine learning. Long Beach, California, USA Contributed Talk: <i>Understanding and controlling memory in recurrent neural networks</i>	2019

TEACHING AND MENTORSHIP

Intern Mentor, Dana Pe'er Lab, Memorial Sloan Kettering Cancer Center Shouvik Mani , Spatial Optimal Transport for analyzing cellular microenvironments Yasa Baig , Discrete latent models for interpretable single-cell analysis	2022 2021
Teaching Assistant, Technion - Israel Institute Of Technology, Introduction to Biological Systems and Signals , Head TA Electromagnetic Fields	2018-2019

REVIEWING

- **Journals:** Nature Biomedical Engineering, Nature Biotechnology, Cell, Genome Biology
- **Conferences:** NeuRIPS, ICLR, ICML, ICML Workshop in Computational Biology